Projects: Generative AI-Based Ligand Binding Pose Refinement from Physics-Based Priors

Literature Review: Scoring Functions for Ligand Binding Pose Prediction

Abstract

The purpose of this literature review is to study the different scoring functions, their use cases for specific proteins or ligands, pros/cons and the method they follow to evaluate the docking pose. After studying the different scoring functions our goal will be to find the best possible scoring method related to our project.

Review

Force Field Scoring Function:

They are centered on molecular mechanical calculations. These scoring functions rely on the fundamental molecular physics terms such as Van der Waals interactions, electrostatic interactions and desolvation energies. These terms can be derived from both experimental data and ab initio quantum mechanical calculations. Programs such as GoldScore, DOCK and early versions of AutoDock use this type of scoring function.

Pros:

1. Each term in the energy equation relates to a clear physical force, making it easy to understand why a molecule scores well or poorly.
2. Provides a natural solution signal to regularize a generative model, as it doesn’t rely on any memorized structural patterns or prior data.

Cons:

1. Computationally more expensive due the solvation and entropy terms.
2. Estimating the loss of conformational freedom when a molecule binds, is very hard and often left out or simplified.
3. Accurately measuring how water molecules leave the binding site remains a major challenge.

Empirical Scoring Function:

They estimate the binding affinity of a complex on the basis of a set of weighted energy terms

$$\Delta G=\sum_{i} W_i * \text{¿} \Delta G_i \text{¿}$$

where ΔG_i represents different energy terms such as VDW energy, electrostatics, hydrogen bond, desolvation, entropy, hydrophobicity, etc. The corresponding coefficients W_i are determined by fitting the binding affinity data of a training set of protein-ligand complexes with known 3D structures. AutoDock Vina is the most widely used example. It evaluates the binding energy using terms for Van der waals forces and hydrogen bonding and runs roughly two orders of magnitude faster than AutoDock4 while beating its accuracy.